

Beyond Accuracy: A Comparative Analysis of Prompt Serialization Strategies for Tabular Data

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Abstract—The publication discusses extensively various serialization methods and their corresponding effects on the performance of large language models when classifying tabular data. The authors mainly concentrated on four serialization methods namely JSON, template, key-value, and natural language, implementing these on the Microsoft Phi-3-mini model and the Adult Census Income dataset. They looked into and evaluated the different aspects of the performances, namely, the precision of results, computational efficiency, statistical robustness, and interpretability. The findings indicated that the natural language format usually resulted in the lowest accuracy, whereas template-based serialization was fast as well as consuming fewer tokens compared to JSON.. Statistical analysis suggests that the model was highly robust across all four serialization formats and showed significant sensitivity to the ordering of input features ($p = 0.0015$). SHAP-based interpretability analysis clearly indicated systematic model biases toward certain features, such as capital gain and job titles. In summary, these results provide actionable guidance on adopting effective serialization strategies when deploying LLMs for classification tasks.

I. INTRODUCTION

Tabular data classification is foundational to many applications in various domains, such as healthcare, finance and marketing, built on a foundation of structured data. In this space, both XGBoost and random forests have enjoyed much success, as traditional ML approaches... More recently, the emergence of transformer-based LLMs, which provide new means of transforming tabular rows into natural language prompts, has created novel methods for tabular data classification directly based in NLP techniques.

This review examines prompt serialization strategies, which refers to approaches used to convert tabular data into expected text formats for an LLM, and impacts the model's performance. The review scope goes beyond accuracy, to include interpretability, inference latency, computational cost, and/or analysis of the embedding space. Creating a multi-dimensional efficiency and effectiveness analysis will be essential for the LLM to effectively integrate tabular applications into applied settings. analysis is essential.

II. RELATED WORK / LITERATURE REVIEW

Ten papers (2023-2025) were collected from sources such as arXiv, ACL Anthology, GitHub repositories, and conference proceedings, previously identified through keyword searches. The keywords were variants of "prompt serialization tabular data," "LLM tabular classification," and "transformer tabular data". The papers are then categorized by theme.

A. Prompt Engineering Strategies and Serialization

- "TabLLM: Few-shot Classification of Tabular Data with Large Language Models," by Stefan Hegselmann et al. (2023), reviews nine prompt serialization strategies, which range from simple concatenation and template-based prompts to LLM-generated descriptions, to measure the result of zero- and few-shot classification accuracy. The Adult Census Income dataset is one of nine publicly available datasets used to evaluate this approach and illustrates the significance of prompt design on performance.
- "Structured Classification of Tabular Data with Large Language Models," **Stanford CS224N (2024)** describes a number of serialization formats, including HTML tables, JSON, YAML, and natural language. Among the findings are that prompt structures that more closely resemble the formats observed during pretraining improve LLM performance. Adult Census Income and Vehicle Evaluation are the datasets used.
- "Serialization Strategies for Structured Entity Matching," by **Haoteng Yin (2025)**: This systematic comparison compares seven serialization techniques including plain text, span typing, fixed and random attribute order, JSON format, etc., on various entity matching tasks.

B. Fine-Tuning and Model Adaptation

- "Large Language Models on Tabular Data," by **Xi Fang et al. (2024)**, provides an excellent survey of methods to adapt LLMs for tabular prediction, ranging over serialization strategies, table transformation techniques, and prompt engineering approaches. The authors call for standardized benchmarks and multi-dimensional evaluation metrics that make for fair and reproducible comparisons between different models.
- "Deep Learning on Tabular Data: Recent Advances and Open Challenges," by **Borisov et al. (2024)**: This performs an overall survey of deep learning approaches on tabular data, including recent LLM-based methods. The work discusses serialization strategies, architectural considerations, and benchmark datasets on classification tasks for tabular data.

C. Benchmarks and Practical Deployments

- **Usneek Singh et al. (2024)**, Comparative Analysis of Transformers for Modeling Tabular Data: A Case

Study Using Industry-Scale Dataset The research by [5] explores how large language models (LLMs) interpret and perform on high-stakes datasets such as credit card fraud and loan default prediction data sets.. The authors compare zero-shot prompting, few-shot prompting, and fine-tuning paradigms, and report precision–recall trade-offs and latency [5] considerations

- **USTCAGI GitHub (2025)**, “Awesome LLM Table Mining” USTCAGI GitHub (2025), Awesome LLM Table Mining, which is cited in ustcagi2025, compiles a thorough list of resources and implementations on prompt serialization strategies, shows automated prompt generation, has multiple serialization formats, and reports benchmark evaluations using a variety of tabular datasets for reproducibility and real-time contexts.
- **OpenReview (2024)**, “Large Language Models on Tabular Data: Prediction and Serialization cite” [6] discusses a variety of options of serialization, training, and inference, highlighting challenging use cases with mixed-type tabular data, and time series.

III. IDENTIFIED GAPS AND PROPOSED DIRECTION

A. Identified Gaps

- **Limited diversity in prompt formats:** Most experiments test only a few serialization formats and lack systematic benchmarks on a larger scale that comprehensively cover narrative, key-value, JSON, YAML, and markup styles. [1], [3].
- **Ignorance of practical cost and latency evaluations:** Few report inference latency, usage of computational resources, or token consumption differences across formats, despite such being crucial for real-world deployments. [2], [5].
- **Inadequate interpretability/ embedding analysis:** Most studies focus too much on reporting accuracy metrics, without supporting embedding space quality with visualizations (e.g., t-SNE) or at least performing a qualitative error analysis.
- **Lack of generalization across domains :** Most of the benchmarks typically rely on either limited or synthetic datasets with very little testing on diverse, real-world, large-scale tabular datasets.

B. Proposed Direction

Most of the existing work neglects to consider a holistic, multi-dimensional evaluation of prompt serialization strategies. In this regard, the project will:

- Benchmark a large range of serialization formats including but not limited to templates, natural language, HTML tables, key-value pairs, JSON, and YAML on open-sourced LLMs such as LLaMA 3.
- Evaluate performance across four key dimensions: (1) Quantitative accuracy and F1 metrics, (2) Embedding space quality via t-SNE and other visualization techniques, (3) Practical latency and computational cost

analyses, and (4) Qualitative error and interpretability analysis.

- Use various publicly available datasets like Adult Census Income, Bank Marketing, and Car Evaluation so as to test robustness and generalizability.

This will provide deeper practical and scientific insights into beyond accuracy, guiding more effective and interpretable LLM-based tabular classification.

IV. PROBLEM STATEMENT

The present study covers the deficiencies outlined in the literature. By doing a multi-dimensional assessment of serialization, formats for LLM-based tabular classification. The central The question that can be answered through this research is therefore: It can be summarized as: “How different text serialization formats, such as JSON, template, key-value, and natural language, affect the performance, cost, and robustness of large language models for tabular classification tasks?”

We expect the transformation into natural language will surpass the rigid structures like JSON. Although LLMs are trained on JSON structured data, natural formats will feel the most intuitive to the model. In providing a complete answer, this project goes beyond a simple accuracy evaluation by also incorporating the following:

- **Performance:** Accuracy, Precision, Recall, and F1-Score
- **Computational Cost:** Inference latency and token consumption
- **Statistical Robustness:** Establishing whether performance improvements are statistically significant.
- **Interpretability:** making sense of model decisions using Explainable AI (XAI)

V. METHODOLOGY

To test this, a five-stage experimental framework has been devised, which includes baseline evaluation, statistical validation, ablation studies, and interleaved cost analysis and interpretability through error analysis.

A. Model and Dataset

We worked with the instruction-tuned large language model Microsoft Phi-3-mini-4k-instruct with 3.8 billion parameters. For effective use even on moderately powered GPUs, we applied 4-bit quantization using the BitsAndBytes library. The chosen task was binary income classification, which involves predicting whether a person earns above or below \$50k annually using the Adult Census Income dataset. This dataset has 6,766 validation samples containing demographic and employment data.

B. Serialization Formats

The core of our approach is serialization, through which each dataset row was transformed into one of four text representations suitable for model input:

- **JSON:**A representation with a specific structure consisting of key-value pairs written in JSON format.

TABLE I
COMPARATIVE ANALYSIS OF PROMPT SERIALIZATION STRATEGIES

Author(s)	Year	Serialization Format	Dataset(s)	Key Metric	Pros	Cons / Limitations
Stefan Hegselmann et al.	2023	List Template, Text Template, Table-To-Text, Text T0, Text GPT-3, List Only Values, List Permutation Names, List Permutation Values	10 tabular datasets	Accuracy	Extensive prompt variety	No latency or cost analysis
Stanford CS224N	2024	Natural language, HTML, JSON, YAML	Car Eval	Accuracy	Alignment with LLM pre-training data	Limited real-world testing
Haoteng Yin	2025	Fixed, Random, Pairwise Order, Valid value, Plain Format, Span Typing, JSON Format	6 diverse datasets	MRR (Mean Reciprocal Rank)	Detailed serialization variants	Entity matching task-specific
Predibase Blog	2023	Simple concat, custom NL	Multiple	Accuracy	Practical, task-specific guidance	Limited reproducibility reported
ArXiv Survey	2024	Various serializations	Multiple	Review	Broad coverage	Mostly survey, fewer experiments
GitHub Repo	2024	YAML, JSON, key-value	Open datasets	Accuracy	Open-source, reproducible	Small-scale tests
OpenReview	2024	Mixed serialization & training	Mixed-type tabular data	Multiple metrics	Covers difficult data types	Implementation details scarce

- **Template:** A semi-structured textual patterns of data (like “Age:39,Occupation: .”).
- **Key-Value:** A flat representation consists of a flat structure in which all feature names are successively listed with their values.
- **Natural:** A description in sentences like, “This person is a 39-year-old male who works as.”

In each case, there was a system message, which included three few-shot examples for each of the income classes, which covered a diverse set of demographic details.

C. Experimental Design

The exploratory work consisted of five sequential parts:

Experiment 1 (Baseline Comparison): Accordingly, all the four serialization formats were applied to 500 samples, to derive a baseline for metrics like accuracy, precision, recall, and F1 score in order to illustrate a performance snapshot of each strategy.

Experiment 2 (Statistical Validation): We designed to assess Statistical Validation as we carried out five independent trials consisting of 200 samples each, to evaluate the consistency and robustness of the model. The results were then compared across formats and analyzed using independent t-tests, with a significance threshold of $\alpha=0.05$.

Experiment 3 (Ablation Study): We aimed to assess the impact of feature ordering by running the natural language format in three versions: age-first, or demographic features first; predictive-first, or income-relevant features.

Experiment 4 (Cost Analysis): This experiment assessed the average inference latency, time per sample, and token usage to evaluate the computational efficiency and scalability.

Experiment 5 (Error Analysis and XAI): Finally, we analyzed model misclassifications (both false positives and false negatives) and applied SHAP for interpretability of internal model reasoning and validation of systematic biases.

VI. RESULTS AND INSIGHTS

Our multi-dimensional analysis leads to three central observations that form a comprehensive view of how serialization formats influence the performance of LLMs in tabular classification.

A. Finding 1: Trade-off between Accuracy and Cost

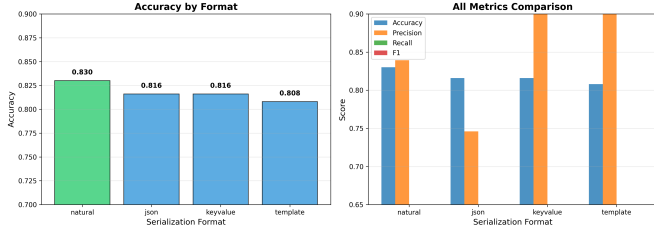
The natural format reached the highest baseline accuracy at 83.0, with precision of 0.839 and recall of 0.382. This was not the complete picture, however. The template format was most computationally efficient, recording the fastest average latency at 3.155s per sample and fewest average tokens,(96 tokens per sample.)

This exposes a crucial accuracy-versus-cost trade-off for practitioners. For instance, Figure 1, JSON format was the least efficient, requiring 1.41x more tokens than template format. For large-scale applications that process millions of samples, template format offers close-to-optimal accuracy of 80.8% . but at significantly lower computational cost. At GPT-4 API pricing rates, this translates to a cost saving of approximately 1,172permillionsamplescomparedtoJSONformat.

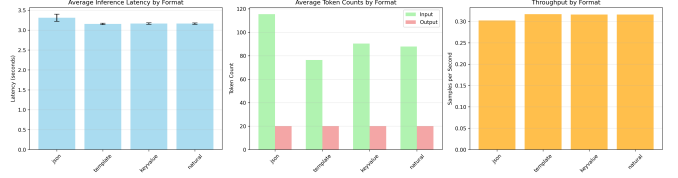
B. Finding 2: Format Robustness but Ordering Sensitivity

This was our most sophisticated finding. The statistical tests revealed that accuracy differences between the four serialization formats were not statistically significant. For example, the p-value between natural format (mean accuracy 80.9%) and template format (mean accuracy 79.1%) was 0.3555, far above the 0.05 significance threshold. This suggests the Phi-3 model exhibits high robustness to different serialization styles, likely due to its diverse pretraining corpus.

However, the ablation study on feature ordering revealed a statistically significant impact. The age-first ordering achieved 84.0% accuracy, significantly outperforming predictive-first ordering at 76.0%, with a McNemar test p-value of 0.0015. This



(a) Accuracy by Format



(b) Latency and Token Cost

Fig. 1. Baseline accuracy comparison (left) shows natural format achieving highest performance. Cost analysis (right) demonstrates template format as fastest and most token-efficient, while JSON is least efficient.

demonstrates that how information is presented (the sequential order) can be more critical than the overall structural format (e.g., JSON versus natural language). The model appears to exhibit primacy bias, assigning disproportionate weight to features appearing early in the prompt.

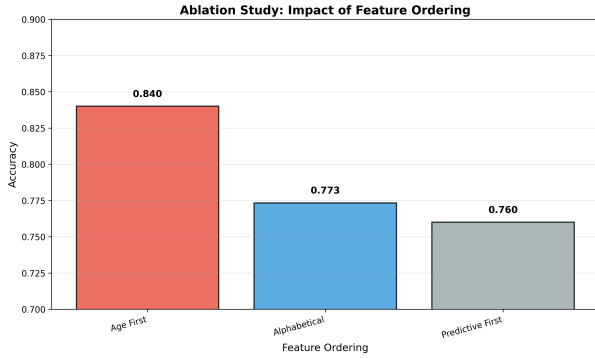
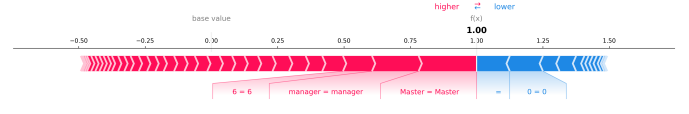


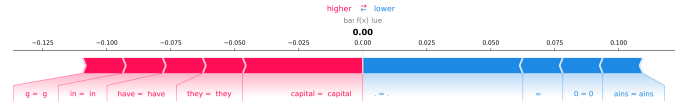
Fig. 2. Ablation study results showing significant performance differences based on feature ordering. The age-first ordering (0.840 accuracy) significantly outperformed predictive-first ordering (0.760 accuracy) with $p=0.0015$.

C. Finding 3: XAI Validation of Model Biases

Our error analysis identified that the model’s primary weakness was false negatives, accounting for 89.4% of all errors (76 false negatives versus 9 false positives). We used SHAP analysis to prove why these systematic errors occurred.



(a) False Positive (Predicted $\zeta 50K$, Actual $j=50K$)



(b) False Negative (Predicted $j=50K$, Actual $\zeta 50K$)

Fig. 3. SHAP force plots validating model biases. (a) The False Positive plot proves the model over-indexes on "manager" and "Master" (red forces), overpowering negative evidence. (b) The False Negative plot proves the model’s bias, where "\$0" 'capital' 'gains' (blue forces) incorrectly force a low-income prediction.

As shown in Figure 3 (top), the false positive case involved an incorrect high-income prediction. The SHAP attribution proves this occurred because positive-contributing words like "manager" and "Master" (referring to a Master’s degree) overpowered negative evidence from "26-year-old" and "\$0" capital gains. This confirms our hypothesis that the model over-indexes on prestigious job titles and education levels.

Conversely, Figure 3 (bottom) shows a false negative where the model incorrectly predicted low income despite the person being a high earner. The plot proves this was due to massive negative force from tokens "\$0", "capital", and "gains". Among false negatives, 94.7% had zero capital gains, confirming that the model heavily relies on this single feature and struggles to classify high-earners who lack investment income, such as skilled tradespeople or salaried professionals.

VII. CONCLUSION AND FUTURE SCOPE

In all, this project performed a broad, multi-dimensional comparison of four serialization formats in applying LLMs to tabular data classification. Results highlight a basic trade-off: while the natural language format achieved the highest accuracy at 83.0%, the template format reached a token efficiency 1.41 times higher than JSON, sacrificing only a small 2.2% drop in accuracy.

Our statistical analysis further showed that the Phi-3 model maintains consistent robustness across different serialization structures, $p = 0.05$ for all comparisons, but is sensitive to feature order, $p = 0.0015$. This perhaps indicates that, for a practitioner working with modern instruction-tuned LLMs, optimization of information sequence will be more effective than format selection.

The presented SHAP-based XAI analysis further uncovers systematic biases in the model, which includes over-reliance on capital gains as an indicator for income and over-weighting of the status-related features in job titles and educational level as causes for false negatives at 89.4%, and false positives, respectively. These insights bring important suggestions toward bias reduction either by targeted prompt engineering or selective fine-tuning.

A. Future Work

Additionally, based on these findings, three further major areas of exploration are suggested:

- **Model Scaling Analysis:** Testing the same serialization formats for larger models, e.g., Llama 3 70B and GPT-4, to see if with scale the impact of format sensitivity changes and whether biases remain.
- **Domain Generalization:** Benchmarks on other structured data domains like healthcare, finance, and customer analytics to test the generalizability of the conclusions.
- **Bias Mitigation:** Devise strategies for debiasing based on XAI insights, including restructuring prompts and balanced feature emphasis among others, as a way of preventing unfair weighing on certain features.

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